

Q1)

1. "unauthorized users attempting to access the system" Unauthorized users **attempting** to access the system shows that there maybe some undiscovered entry points into the network.
2. " website became unavailable due to a Denial of Service (DoS) attack" If a website becomes unavailable due to a DoS attack, it shows that the company's Firewall, IDS, IPS systems are sub optimal. Can incur financial losses.
3. "sensitive customer information was transmitted without encryption" If information is transmitted without encryption, a man in the middle attack could easily edit/steal that data during transit.
4. "some employees accessed confidential company files without permission" If employees can access confidential files without permission, it shows a severe flaw in authorization and access control.
5. "network logs were not regularly monitored." If network logs aren't properly monitored, threat actors can easily get away with their actions as its harder to pin point them.

Q2)

1. Proper Firewall,IDS and IPS solutions can protect the organization against cyber attacks such as DoS.
2. Proper authentication and access controls can ensure only authorized individuals get access to confidential data and access into the system.
3. Proper encryption of data at rest and at transit would preserve data integrity of the organization.
4. Proper monitoring of logs would hold individuals accountable for their actions.

Q3)

- **Confidentiality:** To verify that sensitive data is protected from unauthorized disclosure during transit and storage.
- **Integrity:** To ensure that data cannot be altered or tampered with without detection.
- **Availability:** To evaluate whether network infrastructure is resilient against disruptions (like DoS attacks).
- **Compliance:** To ensure network security practices align with organizational policies and legal regulations.

Q4)

1. Inspect firewall,IDS, IPS configurations.
2. Review access control and authentication policies.
3. Examine the log management policy and logging frequency.
4. Inspect the network configuration to ensure secure protocols and encryption algorithms are being used.

Q5)

1. Install hardware and software firewalls at the network edge.
2. Create a DMZ(demilitarized zone) and keep confidential data further inside the intranet.
3. Use a AAA(triple A) system to provide Authentication, Authorization and Accounting(logging).
4. Use WPA3 enterprise with a RADIUS server for wireless authentication.